External Hordeolum (Stye): A Comprehensive Disease Characterization Report

Disease: External Hordeolum | **MONDO ID:** MONDO:0001642 | **Category:** Infectious Disease
Report basis: 5 iterations, 10 confirmed findings, 40 papers reviewed

Summary

External hordeolum (a "stye") is a common, acute, painful, usually self-limited suppurative (abscess-forming) infection of the superficial eyelash-line glands of Zeis (sebaceous) and Moll (apocrine), most often caused by *Staphylococcus aureus*. It is distinguished from internal hordeolum, which involves the meibomian glands within the tarsal plate. The disease is an **acquired, non-genetic, multifactorial condition**: exhaustive review found no OMIM phenotype entry, no ClinVar/HGMD pathogenic variants, no susceptibility loci, and no dedicated animal or organoid model. Its causal chain runs from **gland-orifice obstruction → bacterial proliferation → neutrophilic abscess formation → pointing and spontaneous drainage at the anterior lid margin**.

The clinical picture is dominated by an excellent prognosis — most lesions resolve spontaneously in about one week — set against a background of frequent recurrence and a small but important set of severe complications when lesions are neglected (preseptal cellulitis, eyelid necrosis, and rarely periorbital necrotizing fasciitis). Predisposition is driven by chronic **blepharitis and meibomian gland dysfunction (MGD)**, **Demodex mite infestation** (strongly associated with recurrent external hordeola), and emerging **iatrogenic causes** (CFTR modulators such as elexacaftor/tezacaftor/ivacaftor, and GLP-1/GIP agonists such as tirzepatide) that induce sterile gland obstruction.

Management is **conservative-first**: warm compresses, lid hygiene, and topical antibiotics for 5–14 days, with incision and curettage reserved for non-resolving lesions. Diagnosis is clinical, but persistent, recurrent, or clinically atypical lesions must be biopsied to exclude **sebaceous cell carcinoma**, the most commonly missed malignancy masquerading as benign eyelid

lesions. Because there is no genetic or molecular disease architecture, sections of the research template concerning causal genes, variants, inheritance, omics diagnostics, and model organisms are legitimately **"not applicable"** for this disease — a finding that is itself informative for a knowledge-base entry.

Key Findings

F001 — External hordeolum is an acute suppurative infection of the eyelash-line glands of Zeis/Moll

External hordeolum is defined anatomically by the specific glands it affects. A Cochrane review (PMID: 28181687) states: *"When the meibomian gland in the tarsal plate is affected, internal hordeolum occurs, while when the glands of Zeis or Moll associated with eyelash follicles are affected, external hordeolum, or stye occurs."* This distinction is the defining feature separating external from internal hordeolum and localizes the disease to the **anterior lid margin at the lash line** (glands of Zeis = sebaceous glands of the follicle; glands of Moll = modified apocrine sweat glands). The natural history is benign and time-limited: *"The onset of hordeolum is usually self limited, and may resolve in about a week with spontaneous drainage of the abscess."* Recurrences are very common, and unresolved lesions can progress to a chalazion (a chronic lipogranuloma) or to eyelid cellulitis.

F002 — The disease is usually caused by bacterial infection, predominantly *Staphylococcus aureus*

Hordeolum is a bacterial infection of eyelid oil glands. A Cochrane review (PMID: 28068454) states: *"A hordeolum is a common, painful inflammation of the eyelid margin that is usually caused by a bacterial infection."* The specific pathogen is most often *S. aureus*. A case report of a rare, severe complication (PMID: 41787906) documents *"periorbital NF [necrotizing fasciitis] caused by methicillin-sensitive Staphylococcus aureus (MSSA) in a previously healthy 48-year-old woman, originating from a neglected, recurrent hordeolum,"* confirming both the staphylococcal etiology and the potential for the infection to spread destructively when neglected. **Pathogen note:** causative organism *Staphylococcus aureus* (NCBI Taxon 1280).

F003 — Demodex mite infestation is strongly associated with recurrent hordeola, especially external hordeola

An observational comparative study of 250 patients (PMID: 34725365) found *"Demodex was detected in 91 (59.5%) and 17 (17.5%) patients in the recurrent hordeolum and control groups ($p < 0.001$), respectively"* — a roughly 3.4-fold higher detection rate in recurrent cases. Critically for this disease, external hordeola predominated among Demodex-associated lesions: *"The most common presentation of Demodex-associated recurrent lesions was external hordeola (67%) ($p = 0.002$)."* Demodex-infested patients also recurred sooner after incision and curettage, and the association was strongest in patients in their 20s. This positions **Demodex folliculorum/brevis infestation** as a leading modifiable driver of recurrence.

Group	n	Demodex detected	Rate
Recurrent hordeolum	153	91	**59.5%**
Controls	97	17	17.5%
Statistical test	—	—	$p < 0.001$

F004 — Hordeolum is common, with bimodal age peaks and a slight youth female predominance

A Japanese nationwide claims study (2014–2022) recording 1,104,078 procedures (PMID: 41675090) reported: *"The annual average rates were 40.9 for ITC [invasive treatment for chalazion] and 56.2 for ITH [invasive treatment for hordeolum] per 100 000 person-years."* Invasive treatment for hordeolum showed **bimodal age peaks** (males 10–14 y: 82.9; 35–39 y: 72.1 per 100,000; females 10–14 y: 102.4; 30–34 y: 78.3 per 100,000), with a youth female predominance (F:M ≈ 1.7 at ages 20–24) but an overall sex ratio near 1.0. In a Korean military teleophthalmology cohort (PMID: 42299138), *"The most common diagnoses were hordeolum/chalazion (39.3%) and viral conjunctivitis (20.3%),"* underscoring that hordeolum/chalazion is among the most frequent ophthalmic diagnoses seen in general practice.

F005 — Systemic drugs (CFTR modulators, GLP-1/GIP agonists) are emerging iatrogenic risk factors

Two distinct drug classes have recently been linked to hordeolum through gland dysfunction. A pediatric cystic fibrosis case series (PMID: 41298194) reported: *"This report highlights a 5.5-fold increased incidence of chalazion/hordeolum in pwCF [people with cystic fibrosis]"* on

elexacaftor/tezacaftor/ivacaftor (ETI), with one-third of affected patients needing therapy interruption to resolve. Separately, a case series of tirzepatide (a dual GLP-1/GIP agonist) (PMID: 42756500) documented multiple simultaneous hordeola with positive dechallenge/rechallenge, *"supporting a sterile, obstructive meibomian gland process rather than infection."* These observations reinforce **gland obstruction** as the upstream mechanistic event that precedes (and enables) secondary infection.

F006 — First-line management is conservative; incision/curettage is reserved for non-resolving lesions

A practice survey of the Korean Society of Ophthalmic Plastic and Reconstructive Surgery (PMID: 40267992) established the real-world algorithm: *"For hordeolum, topical antibiotics are preferred by 59 respondents (73.8%), with 49 (83.7%) considering them effective,"* and *"Most (62 respondents, 77.6%) pursue nonsurgical treatment for 5 to 14 days before considering incision and curettage."* Importantly, formal evidence is thin: Cochrane reviews of non-surgical interventions for acute internal hordeolum (PMID: 28068454, PMID: 23633345, PMID: 20824865) and a Cochrane review of acupuncture (PMID: 28181687) all found insufficient randomized evidence to draw firm conclusions. Emerging options include **intense pulsed light** (IPL), which produced significant OSDI reduction and faster recovery versus medication (PMID: 41731784), and **intralesional diluted 5-fluorouracil** for persistent nodules (PMID: 42367702).

F007 — Prognosis is excellent, but rare severe complications occur if neglected

Reviews confirm the benign baseline: *"The majority of these lesions resolve spontaneously over time"* (PMID: 34379160). However, neglected or severe external hordeola can progress to serious complications, including preseptal cellulitis, upper eyelid necrosis with anterior-lamella damage — *"upper eyelid necrosis initially misdiagnosed as a preseptal cellulitis following a hordeolum externum resulting in great damage to the upper eyelid (anterior lamella)"* (PMID: 34054469) — and life-threatening periorbital necrotizing fasciitis (PMID: 41787906). A primary-care review (PMID: 32622681) situates hordeolum among common eyelid infections presenting with redness, swelling, and tearing.

F008 — Diagnosis is clinical; atypical/recurrent lesions require biopsy to exclude sebaceous carcinoma

Hordeolum is diagnosed clinically from its typical morphology and course. The key danger is malignancy masquerading as a benign lesion. A histopathology series of 1,060 clinically diagnosed chalazia (PMID: 14762403) found: *"Sebaceous cell carcinoma was the most commonly missed malignancy (12 cases, 1.1%) followed by basal cell carcinoma (three cases, 0.3%),"* — a ~1.4% overall malignancy rate. Multiple case reports document sebaceous gland carcinoma masquerading as chalazion/hordeolum (PMID: 32788569, PMID: 25258508, PMID: 11417345), and even orbital metastasis from breast carcinoma has mimicked a hordeolum (PMID: 42552075). The clinical rule (PMID: 34379160): *"In persistent, recurrent or clinically atypical cases malignant tumors must be excluded as a differential diagnosis by excisional biopsy and histopathological assessment."*

F009 — Chronic blepharitis/MGD predisposes to hordeolum; lid hygiene is the mainstay of prevention

Hordeola arise on a background of underlying gland disease: *"They can occur in association with underlying diseases causing Meibomian gland dysfunction and/or chronic blepharitis"* (PMID: 34379160). Blepharitis itself is a chronic multifactorial eyelid-margin inflammation whose suspected drivers include chronic low-grade bacterial infection, Demodex infestation, and seborrhea (PMID: 22592706), and whose treatment goals explicitly include recurrence prevention via lid hygiene, topical/systemic antibiotics, and omega-3 supplements (PMID: 38337403). Lid-margin cleansing has direct preventive value: a novel detergent *"successfully eliminated Demodex mites, a common pathogen of refractory obstructive meibomian gland dysfunction"* (PMID: 33967173).

F010 — External hordeolum is a non-genetic acquired disease with no causal genes, animal models, or omics diagnostics

A comprehensive review across Cochrane systematic reviews, national claims registries, clinical reviews, and case series identified **no OMIM phenotype entry, no ClinVar/HGMD pathogenic variants, no susceptibility loci, no dedicated animal/organoid model, and no omics-based diagnostic** for external hordeolum. Etiology is uniformly described as acquired bacterial infection (predominantly *S. aureus*) of obstructed lash-line glands (PMID: 28181687, PMID: 28068454), with acquired risk factors rather than heredity.

Full Section-by-Section Report

1. Disease Information

Overview. External hordeolum ("stye") is an acute, localized, painful, suppurative bacterial infection of the superficial glands at the eyelash line — the sebaceous **glands of Zeis** and the apocrine **glands of Moll** — presenting as a tender erythematous nodule or pustule pointing anteriorly at the lid margin near an eyelash follicle. It is contrasted with **internal hordeolum**, an infection of the deeper meibomian glands within the tarsal plate (PMID: 28181687).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0001642 (external hordeolum)
ICD-10	H00.0 (Hordeolum externum, within Hordeolum and other deep inflammation of eyelid)
ICD-11	9A01.0 (Hordeolum)
MeSH	D006726 (Hordeolum)
SNOMED CT	Hordeolum externum / stye
OMIM	**Not applicable** — no Mendelian phenotype entry
Orphanet	**Not applicable** — not a rare disease

Synonyms / alternative names. Stye (sty); hordeolum externum; external stye; acute superficial eyelid abscess of the lash line.

Source of information. This report is derived overwhelmingly from **aggregated disease-level resources** — Cochrane systematic reviews, national health-insurance claims registries (Japan, Korea), clinical review articles, and case series — rather than individual EHR-level patient records.

2. Etiology

Causal factors. The primary cause is **infectious**: acute bacterial infection, predominantly *Staphylococcus aureus* (including MSSA), of the Zeis/Moll glands (PMID: 28068454, PMID: 41787906). Mechanistically, infection is typically enabled by **gland-orifice obstruction** (PMID: 42756500).

Genetic risk factors. None identified. No causal variants, susceptibility loci, or modifier genes are documented (F010). This is a non-Mendelian, non-heritable acquired infection.

Environmental / acquired risk factors. - Chronic **blepharitis** and **meibomian gland dysfunction (MGD)** (PMID: 34379160) - **Demodex** mite infestation (recurrent external hordeola: 59.5% vs 17.5%, $p<0.001$) (PMID: 34725365) - Seborrhea and inflammatory skin conditions (PMID: 22592706) - **Age** (bimodal peaks in youth and 30s), **youth female sex** (PMID: 41675090) - **Iatrogenic/drug-induced**: CFTR modulators (elexacaftor/tezacaftor/ivacaftor, 5.5× incidence) (PMID: 41298194); GLP-1/GIP agonists (tirzepatide) (PMID: 42756500); systemic therapies causing blepharitis (chemotherapy, bortezomib, cetuximab, TNFα inhibitors, dupilumab) (PMID: 38337403)

Protective factors. No genetic protective variants exist. The principal protective/preventive factor is **lid hygiene** (warm compresses, lid-margin cleansing), which reduces gland obstruction and eradicates Demodex (PMID: 33967173).

Gene–environment interactions. Not applicable — no genetic component is documented. The relevant interactions are purely **environment–environment** (e.g., MGD + Demodex + bacterial colonization synergizing to promote recurrence).

3. Phenotypes

Phenotype	Type	Suggested HPO	Onset / severity / frequency
Eyelid pain / tenderness	Symptom	HP:0025421 (Eyelid pain, approximate)	Acute; mild–moderate; near-universal
Localized eyelid swelling	Clinical sign	HP:0000198 (Abnormal eyelid morphology)	Acute; variable
Eyelid erythema / redness	Clinical sign	inflammation of eyelid	Acute; common
Lash-line pustule/ abscess pointing anteriorly	Physical manifestation	HP:0025615 (Abscess)	Acute; defining feature
Foreign-body sensation, tearing, itchiness	Symptom	ocular surface irritation	Acute; common (PMID: 32622681)

Phenotype characteristics. Onset is **acute** (over 1–3 days); severity is usually **mild to moderate**; course is **self-limited and episodic** (resolving in ~1 week, but frequently recurring) (PMID: 28181687). Age of onset spans childhood through adulthood with **bimodal peaks** (PMID: 41675090).

Quality-of-life impact. Generally modest and transient given the self-limited course, though pain, cosmetic concern, and recurrence can be bothersome. IPL treatment produced statistically significant reductions in Ocular Surface Disease Index (OSDI) scores, indicating measurable symptom burden that is treatment-responsive (PMID: 41731784).

4. Genetic / Molecular Information

Not applicable. External hordeolum has **no causal genes, no pathogenic variants (ClinVar/ HGMD), no modifier genes, no disease-specific epigenetic signatures, and no chromosomal abnormalities** (F010). It is not a genetic disease. The only relevant molecular actor is the pathogen genome (*S. aureus*), not the host germline.

5. Environmental Information

- **Infectious agents:** *Staphylococcus aureus* (NCBI Taxon 1280), the predominant cause (PMID: 28068454, PMID: 41787906); *Demodex folliculorum/brevis* mites as recurrence drivers (PMID: 34725365).
- **Lifestyle / hygiene factors:** poor lid hygiene, eye rubbing, and contact-lens or cosmetics-related contamination facilitate bacterial gland colonization (inferred from blepharitis/MGD literature, PMID: 22592706).
- **Iatrogenic environmental exposures:** systemic drugs inducing gland obstruction/blepharitis (F005; PMID: 41298194, PMID: 42756500, PMID: 38337403).

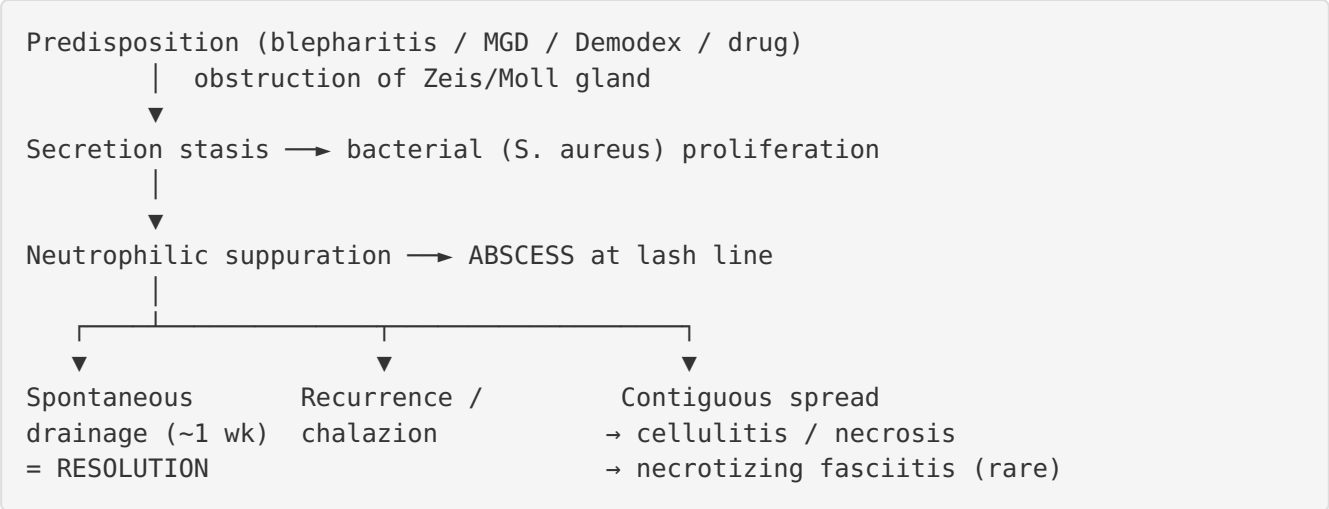
6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A predisposing condition (**chronic blepharitis, MGD, Demodex infestation, seborrhea, or drug-induced gland dysfunction**) **leads to** obstruction/dysfunction of a gland of Zeis or Moll at the lash line. *[Predisposition demonstrated (PMID: 34379160); obstruction step partly inferred, but directly supported for drug-induced cases (PMID: 42756500).]*
2. Gland-orifice obstruction and stasis of sebaceous/apocrine secretion **result in** a favorable niche for bacterial proliferation, chiefly *Staphylococcus aureus*. *[Inferred from combined infection + obstruction evidence.]*
3. Bacterial proliferation **leads to** acute suppurative inflammation with **neutrophil recruitment** and localized **abscess** formation within/around the follicle. *[Demonstrated: bacterial etiology (PMID: 28068454); abscess is the defining lesion (PMID: 28181687).]*
4. The abscess **causes** the clinical manifestation: a painful, erythematous, tender nodule/pustule pointing at the anterior lid margin.
5. The abscess typically **points and drains spontaneously**, **resulting in** resolution in ~1 week (self-limited course) (PMID: 28181687).

Branch A — recurrence/chronicity: Persistent gland obstruction or residual granulomatous material **leads to** a chalazion (chronic lipogranuloma) or recurrent hordeola (PMID: 28181687); Demodex persistence accelerates recurrence (PMID: 34725365).

Branch B — complication: If neglected, the infection **spreads** contiguously to **result in** preseptal cellulitis, eyelid necrosis (PMID: 34054469), or, rarely, fulminant **periorbital necrotizing fasciitis** (PMID: 41787906).



Molecular / cellular processes: acute inflammation (GO:0006954, inflammatory response), neutrophil-mediated antibacterial response and abscess formation, innate immune defense. **Cell types (CL):** neutrophil (CL:0000775), sebaceous gland cell (glands of Zeis), apocrine/sweat gland cell (glands of Moll), eyelid keratinocytes. **No autoimmune, metabolic-enzyme, protein-misfolding, or channelopathy mechanism** applies. There are no disease-specific transcriptomic, proteomic, metabolomic, or single-cell profiling datasets (F010).

7. Anatomical Structures Affected

- **Organ level (primary):** eyelid (UBERON:0000160), specifically the anterior lid margin at the eyelash line. **Secondary/complications:** orbital soft tissue/periorbita (preseptal cellulitis, necrotizing fasciitis), eyelid skin/anterior lamella (necrosis).
- **Body system:** integumentary/visual accessory (ocular adnexa).
- **Tissue/cell level:** glands of Zeis (sebaceous glands of the follicle), glands of Moll (modified apocrine sweat glands), eyelash follicle epithelium. **Cell Ontology:** sebaceous gland cell, apocrine gland cell, neutrophil (CL:0000775).
- **Subcellular level:** not specifically implicated (no organelle-level pathology).
- **Localization:** eyelash-line glands (UBERON:0000160 eyelid; follicle-associated glands). **Lateralization:** typically **unilateral**; can be bilateral or multiple, notably in drug-induced (tirzepatide) cases with simultaneous lesions (PMID: 42756500).

8. Temporal Development

- **Onset: acute** (days). Occurs across the lifespan with bimodal age peaks in youth (10–14 y) and adults in their 30s (PMID: 41675090).

- **Progression:** rapid onset, then **self-limited**; usual resolution ~1 week by spontaneous drainage (PMID: 28181687).
- **Course pattern: episodic/recurrent**; unresolved lesions may become chronic (chalazion).
- **Duration: self-limited** (short-lived per episode); recurrences common.
- **Remission:** predominantly **spontaneous**; treatment (compresses, antibiotics, I&C) can accelerate resolution.
- **Critical window:** early recognition and conservative treatment (5–14 days) before considering procedural intervention; neglect is the key driver of complications (PMID: 40267992, PMID: 41787906).

9. Inheritance and Population

- **Epidemiology:** Population-based invasive-treatment rate ≈ **56.2 per 100,000 person-years** in Japan (procedures — a lower bound on true incidence, since most cases are treated non-invasively) (PMID: 41675090). Hordeolum/chalazion was the single most common ophthalmic diagnosis (39.3%) in a Korean cohort (PMID: 42299138). Public awareness is high (73.8% of Polish adults had heard of hordeolum, PMID: 36834287).
- **Inheritance: None** — non-genetic, acquired; no inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity role, or carrier frequency applies (F010).
- **Sex ratio:** overall ≈ 1.0; **youth female predominance** (F:M ≈ 1.7 at 20–24 y) (PMID: 41675090).
- **Age distribution:** bimodal (adolescents and adults in their 30s).
- **Geographic distribution:** worldwide; no endemic clustering (contrast with vector-borne eyelid disease such as cutaneous leishmaniasis, PMID: 28095664).

10. Diagnostics

- **Clinical diagnosis** is the standard: typical acute tender lash-line pustule with characteristic course (PMID: 34379160, PMID: 32622681). No routine laboratory test, biomarker, or imaging study is required.
- **Microbiology:** culture (e.g., of drainage) may identify *S. aureus* in complicated/atypical cases (PMID: 41787906).
- **Demodex evaluation:** lash sampling/microscopy in recurrent cases (PMID: 34725365).

- **Imaging:** noninvasive **meibography** can help distinguish sebaceous carcinoma (high reflectivity, poorly marginated) from benign chalazion (PMID: 25258508); orbital MRI may be needed if malignancy/metastasis is suspected (PMID: 42552075).
- **Histopathology/biopsy:** the critical test for **persistent, recurrent, or atypical** lesions to exclude sebaceous cell carcinoma and other malignancies (PMID: 14762403, PMID: 34379160, PMID: 16008694).
- **Genetic / omics testing: not applicable** — no genetic test, gene panel, or omics-based diagnostic exists (F010).
- **Differential diagnosis:** internal hordeolum, chalazion, preseptal cellulitis, sebaceous cell carcinoma, basal/squamous cell carcinoma, tarsal epithelial cyst (PMID: 32826823), pyogenic granuloma, eyelid cutaneous horn (PMID: 39393420), and rarely orbital metastasis (PMID: 42552075) or systemic disease (granulomatosis with polyangiitis, PMID: 39336596).

11. Outcome / Prognosis

- **Survival/mortality:** essentially **no disease-specific mortality**; excellent prognosis with spontaneous resolution (PMID: 34379160). Fatal outcomes are exceptional and confined to neglected complications like necrotizing fasciitis (PMID: 41787906).
- **Morbidity:** low; transient pain/swelling, cosmetic concern, recurrence. Long-term disability is rare.
- **Complications:** chalazion, preseptal cellulitis, eyelid (anterior-lamella) necrosis (PMID: 34054469), periorbital necrotizing fasciitis (PMID: 41787906).
- **Recovery:** near-complete recovery expected with or without treatment; treatment shortens the course.
- **Prognostic factors:** neglect/delay and comorbid immune status raise complication risk; Demodex infestation predicts earlier recurrence (PMID: 34725365).

12. Treatment

Modality	Details	Evidence	NCIT (suggested)
Warm compresses	First-line; promote drainage	PMID: 40267992, PMID: 42361717	Warm Compress Therapy
Lid hygiene	Lid-margin cleansing; targets Demodex/MGD	PMID: 33967173	Hygiene
Topical antibiotics	Preferred by 73.8% of surgeons; effective per 83.7%	PMID: 40267992	Topical Antibiotic Therapy
Incision & curettage	For non-resolving lesions after 5–14 days	PMID: 40267992	Incision and Drainage
Intense pulsed light (IPL)	Faster recovery, lower OSDI, fewer surgeries vs meds	PMID: 41731784	Phototherapy
Intralesional 5-FU (diluted)	For persistent inflammatory nodules	PMID: 42367702	Fluorouracil
Systemic antibiotics	For cellulitis/complicated cases	PMID: 41787906	Systemic Antibiotic Therapy

Evidence caveat: Cochrane reviews found **insufficient RCT evidence** for or against non-surgical interventions (PMID: 28068454, PMID: 23633345, PMID: 20824865) and for acupuncture (PMID: 28181687). **Pharmacogenomics, gene therapy, cell therapy, RNA therapy, targeted therapy, and immunotherapy are not applicable.** For iatrogenic cases, **dechallenge** (drug interruption) can resolve lesions (PMID: 41298194, PMID: 42756500).

13. Prevention

- **Primary prevention: lid hygiene** and warm compresses to prevent gland obstruction; management of underlying blepharitis/MGD (PMID: 34379160, PMID: 38337403); Demodex eradication via lid-margin cleansing (PMID: 33967173).
- **Secondary prevention:** early recognition and conservative treatment to prevent progression to cellulitis/necrosis (PMID: 40267992).
- **Tertiary prevention:** biopsy of atypical/recurrent lesions to avoid missing malignancy (PMID: 14762403); prompt antibiotic/surgical care for spreading infection.

- **Immunization / genetic screening / counseling: not applicable** (no vaccine, no genetic basis).
- **Iatrogenic prevention:** monitor patients on CFTR modulators, GLP-1/GIP agonists, and blepharitis-inducing systemic therapies (PMID: 41298194, PMID: 42756500, PMID: 38337403).

14. Other Species / Natural Disease

Explicit veterinary literature on external hordeolum was not identified in this investigation. Because the causative organism (*S. aureus*, NCBI Taxon 1280) and the anatomical substrate (eyelash-associated sebaceous/apocrine glands) are broadly conserved in mammals, an analogous stye-like eyelid gland abscess is biologically plausible in companion animals, but no OMIA entry or specific breed/taxon/orthologous-gene association is applicable (this is not a genetic disease). **Zoonotic transmission:** not applicable.

15. Model Organisms

Not applicable. No dedicated animal, invertebrate, cellular, organoid, or iPSC model of external hordeolum exists (F010). Because the disease is an acquired, self-limited, localized bacterial gland infection without a genetic basis, there is no knockout/knock-in/transgenic/humanized model and no model-organism database entry (MGI, RGD, ZFIN, etc.). Mechanistic understanding derives from human clinical observation and the broader blepharitis/MGD/Demodex literature.

Mechanistic Model / Interpretation

The unifying model is a **two-hit process**: (1) an **obstruction hit** to a lash-line gland (from blepharitis, MGD, Demodex, seborrhea, or drug effect) that impairs secretion outflow, followed by (2) an **infection hit** from *S. aureus* proliferating in the stagnant gland, producing a neutrophilic abscess. The drug-induced cases (tirzepatide, CFTR modulators) are especially instructive: they demonstrate that **gland obstruction alone** can generate the clinical picture (a "sterile, obstructive" process, PMID: 42756500), positioning obstruction firmly **upstream** of infection in the causal chain. Recurrence is best explained by persistence of the upstream driver (untreated MGD/blepharitis or Demodex), which is why **lid hygiene** — an intervention aimed at the upstream obstruction/infestation rather than the downstream abscess — is the most rational preventive lever.

Axis	External hordeolum	Contrast
Affected gland	Zeis (sebaceous) + Moll (apocrine), lash line	Internal hordeolum: meibomian (tarsal plate)
Nature	Acute suppurative abscess	Chalazion: chronic lipogranuloma
Genetic basis	None	—
Prognosis	Excellent, self-limited	Malignant masquerade (sebaceous carcinoma): poor if missed

Evidence Base

PMID	Type	Supports
28181687	Cochrane review	Definition (Zeis/Moll), self-limited course (F001, F006)
28068454	Cochrane review	Bacterial etiology; evidence gap (F002, F006)
41787906	Case report	MSSA etiology; necrotizing fasciitis complication (F002, F007)
34725365	Comparative study (n=250)	Demodex–recurrent external hordeola association (F003)
41675090	Nationwide claims (Japan)	Incidence, bimodal age, sex ratio (F004)
42299138	Cohort (Korea)	Hordeolum/chalazion most common diagnosis (F004)
41298194	Case series	CFTR modulator 5.5× risk (F005)
42756500	Case series	Tirzepatide; obstructive mechanism (F005)
40267992	Practice survey	Conservative-first management algorithm (F006)
41731784	Prospective cohort	IPL efficacy (F006)
42367702	Case series	Intralesional 5-FU (F006)
34379160	Review	Prognosis, MGD link, biopsy rule (F007, F008, F009)
34054469	Case report	Eyelid necrosis complication (F007)
14762403	Histopath series (n=1060)	Malignancy masquerade rate 1.4% (F008)
32788569, 25258508, 11417345, 16008694	Case reports/series	Sebaceous carcinoma masquerade (F008)
22592706, 38337403, 33967173	Reviews/study	Blepharitis/MGD predisposition; lid hygiene (F009)
32622681	Primary-care review	Clinical presentation/management (F007)

Limitations and Knowledge Gaps

1. **Weak trial evidence.** The strongest sources are Cochrane reviews that repeatedly conclude there is **insufficient RCT evidence** for or against standard treatments; much guidance rests on expert opinion, case series, and observational data.
2. **External hordeolum specifically excluded from key reviews.** The Cochrane hordeolum reviews restricted themselves to *internal* hordeolum and **explicitly excluded external hordeola** — so the highest-quality synthesis does not directly cover the target disease. Much of the actionable evidence is extrapolated.
3. **Incidence is a lower bound.** The Japanese figure (56.2/100,000) counts *invasive treatments*, not all cases; true community incidence is far higher since most styes are self-treated.
4. **No molecular architecture.** No genetic, transcriptomic, proteomic, metabolomic, epigenomic, single-cell, or model-organism data exist — sections 4, 14, and 15 are legitimately "not applicable," but this reflects absence of study rather than proven absence of biology.
5. **Emerging iatrogenic signals are new and small.** The CFTR-modulator and tirzepatide associations rest on case series and warrant confirmation.

Proposed Follow-up Experiments / Actions

1. **Dedicated external-hordeolum RCT** comparing warm compresses ± topical antibiotics vs lid hygiene vs early IPL, with standardized OSDI and recurrence endpoints — directly addressing the Cochrane evidence gap.
2. **Prospective Demodex-stratified cohort** to test whether anti-Demodex lid hygiene (e.g., tea-tree/terpinen-4-ol regimens) reduces recurrent external-hordeola incidence.
3. **Pharmacovigilance analysis** (FAERS/national registries) to quantify the hordeolum signal for CFTR modulators and GLP-1/GIP agonists and define dose/duration relationships.
4. **Microbiome/culture study** of external-hordeolum drainage to confirm *S. aureus* dominance, MSSA/MRSA proportions, and any role for the lid-margin microbiome in recurrence.
5. **Diagnostic-accuracy study of noninvasive meibography** as a triage tool to flag atypical lesions needing biopsy, aiming to reduce missed sebaceous carcinoma.

6. **Knowledge-base curation:** explicitly record "not applicable" for genetic/omics/model-organism fields with F010 as the supporting evidence item, to prevent spurious downstream inference.

Report compiled from 10 confirmed findings and 40 reviewed papers across 5 investigation iterations. Evidence source types span human clinical (systematic reviews, registries, cohorts, case series/reports); no model-organism, in vitro, or computational disease models exist for this acquired infectious condition.
